Year 12 Further Maths - Product-moment correlation

Similar questions - Hypothesis tests and interpretation

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A teacher believes that the number of hours a student spends revising and the student's score in a test are positively related. She asks 18 randomly chosen students for the number of hours of revision, x, and their test score, y (as a percentage). The results are summarised as follows.

$$n = 18, \quad \sum x = 216, \quad \sum x^2 = 2916, \quad \sum y = 1170, \quad \sum y^2 = 78300, \quad \sum xy = 14510$$

(a) Calculate the value of Pearson's product-moment correlation coefficient r. **[2]**

(b) State an assumption needed to be able to carry out a significance test on the value of r. **[1]**

(c) Assume that the assumption in part (b) is valid. Test at the 5% significance level whether there is evidence that more revision is associated with higher scores. **[4]**

(d) The teacher decides to record the revision time in minutes rather than hours. State, with a reason, the effect this has on the value of r. **[1]**

2 Fourteen candidates took a mechanics paper and a statistics paper.

(a) For all fourteen candidates, the value of the product-moment correlation coefficient r for the marks on the two papers was 0.571 correct to 3 significant figures. Test, at the 5% significance level, whether there is evidence of association between the marks on the two papers. **[4]**

(b) A teacher decided to omit the candidates who were in the top three places in mechanics and the candidates who were in the bottom three places in mechanics. For the remaining 8 candidates the marks x (mechanics) and y (statistics) are summarised by

$$n = 8, \quad \sum x = 480, \quad \sum y = 536, \quad \sum x^2 = 29100, \quad \sum y^2 = 36900, \quad \sum xy = 32269.$$

(i) Calculate the value of r for these 8 candidates. **[2]**

(ii) What do the two values of r, in parts (a) and (b)(i), tell you about the scores of the fourteen candidates? **[1]**

3 A shopper estimates the cost, £x per item, of each of 10 items in a supermarket. The estimates are compared with the actual cost, £y per item, of each item. The results are summarised as follows.

$$n = 10, \quad \sum x = 28.5, \quad \sum y = 28.45, \quad \sum x^2 = 100.965, \quad \sum y^2 = 94.4375, \quad \sum xy = 95.70$$

Test at the 1% significance level whether the shopper's estimates are positively correlated with the actual cost of the items. **[7]**

4 A psychologist investigates the relationship between 'openness' and 'creativity' in adults. Each member of a random sample of 12 adults is given two tests, one on openness and one on creativity. Each test has a maximum score of 50. The results are given in the table.

Adult	A	B	C	D	E	F	G	H	I	J	K	L
Openness, x	22	35	28	41	19	33	26	38	30	24	44	17
Creativity, y	30	31	22	40	25	36	20	29	38	27	41	24

$n = 12, \sum x = 357, \sum y = 363, \sum x^2 = 11445, \sum y^2 = 11537, \sum xy = 11286.$

(a) Use Pearson's product-moment correlation coefficient to test, at the 5% significance level, whether there is positive association between openness and creativity. **[6]**

(b) State what the value of the product-moment correlation coefficient shows about a scatter diagram illustrating the data. **[2]**

(c) A student suggests that it would be better to standardise the test scores, so that they have the same mean and variance, before carrying out the test. Explain whether you agree. **[1]**

Question 1

(a) $S_{xx} = 2916 - \dfrac{216^2}{18} = 324$; $S_{yy} = 78300 - \dfrac{1170^2}{18} = 2250$; $S_{xy} = 14510 - \dfrac{216 \times 1170}{18} = 470$. $r = \dfrac{470}{\sqrt{324 \times 2250}} = 0.550$

(b) The data come from a bivariate normal distribution (or: both variables are random and approximately normally distributed)

(c) $H_0 : \rho = 0$, $H_1 : \rho > 0$. For $n = 18$ the 5% one-tailed critical value is 0.4000. $0.550 > 0.4000$, so reject H_0: there is evidence at the 5% level of positive association between revision hours and score

(d) No effect, since r is unchanged by a linear change of units

Question 2

(a) $H_0 : \rho = 0$, $H_1 : \rho \neq 0$. For $n = 14$ the 5% two-tailed critical value is 0.5324. $0.571 > 0.5324$, so reject H_0: there is evidence of association between the marks

(b)(i) $S_{xx} = 29100 - \dfrac{480^2}{8} = 300$; $S_{yy} = 36900 - \dfrac{536^2}{8} = 988$; $S_{xy} = 32269 - \dfrac{480 \times 536}{8} = 109$. $r = \dfrac{109}{\sqrt{300 \times 988}} = 0.200$

(ii) The correlation is much weaker when the range of mechanics marks is restricted (omitting the extreme candidates), i.e. the strong overall correlation is largely due to the extreme candidates

Question 3

$H_0 : \rho = 0, H_1 : \rho > 0$. $S_{xx} = 100.965 - \dfrac{28.5^2}{10} = 19.74$; $S_{yy} = 94.4375 - \dfrac{28.45^2}{10} = 13.4973$; $S_{xy} = 95.70 - \dfrac{28.5 \times 28.45}{10} = 14.6175$. $r = \dfrac{14.6175}{\sqrt{19.74 \times 13.4973}} = 0.896$. Critical value ($n = 10$, 1% one-tailed) $= 0.7155$. $0.896 > 0.7155$, so reject H_0: there is evidence at the 1% level that the estimates are positively correlated with the actual costs

Question 4

(a) $S_{xx} = 11445 - \dfrac{357^2}{12} = 824.25$; $S_{yy} = 11537 - \dfrac{363^2}{12} = 556.25$; $S_{xy} = 11286 - \dfrac{357 \times 363}{12} = 486.75$. $r = \dfrac{486.75}{\sqrt{824.25 \times 556.25}} = 0.719$. $H_0 : \rho = 0$, $H_1 : \rho > 0$; critical value ($n = 12$, 5% one-tailed) $= 0.4973$. $0.719 > 0.4973$ so reject H_0: there is evidence of positive association

(b) The points lie reasonably close to a straight line with positive gradient (moderately strong positive linear correlation)

(c) Disagree: r is unchanged by linear scaling of either variable (standardising is a linear transformation), so the test would give the same result